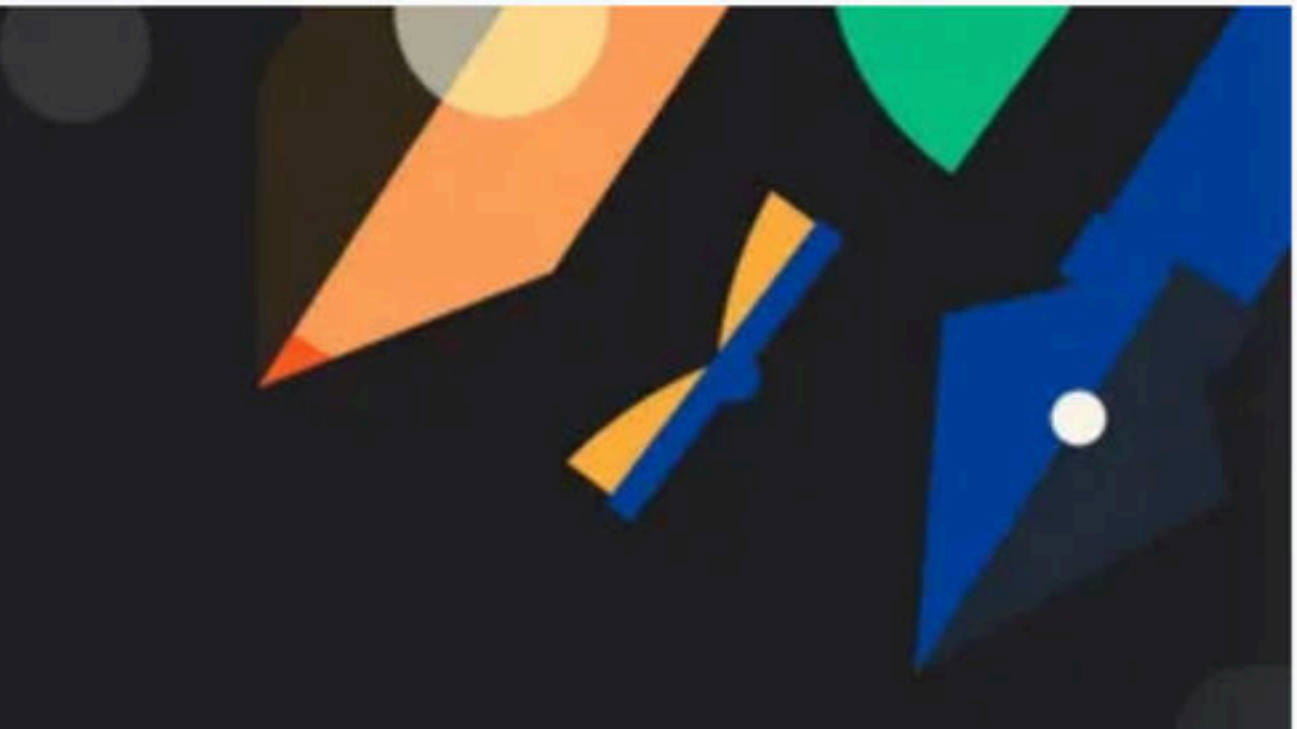




Mega Class: Binary Tree

Special class

Simple → enough → Traversal pakad

Tab tak apki placement na
ho

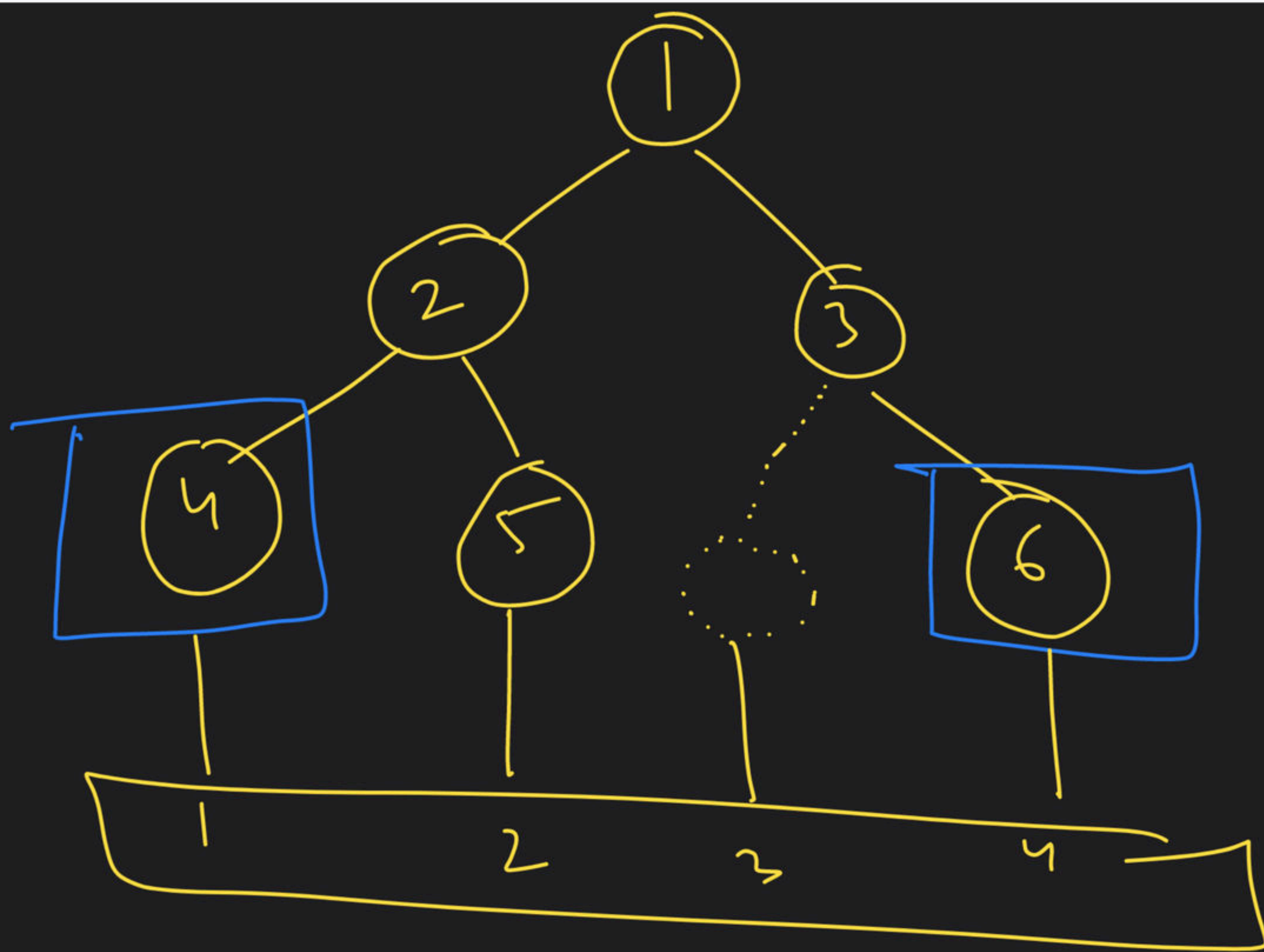
↓
don't stop practicing
DSA

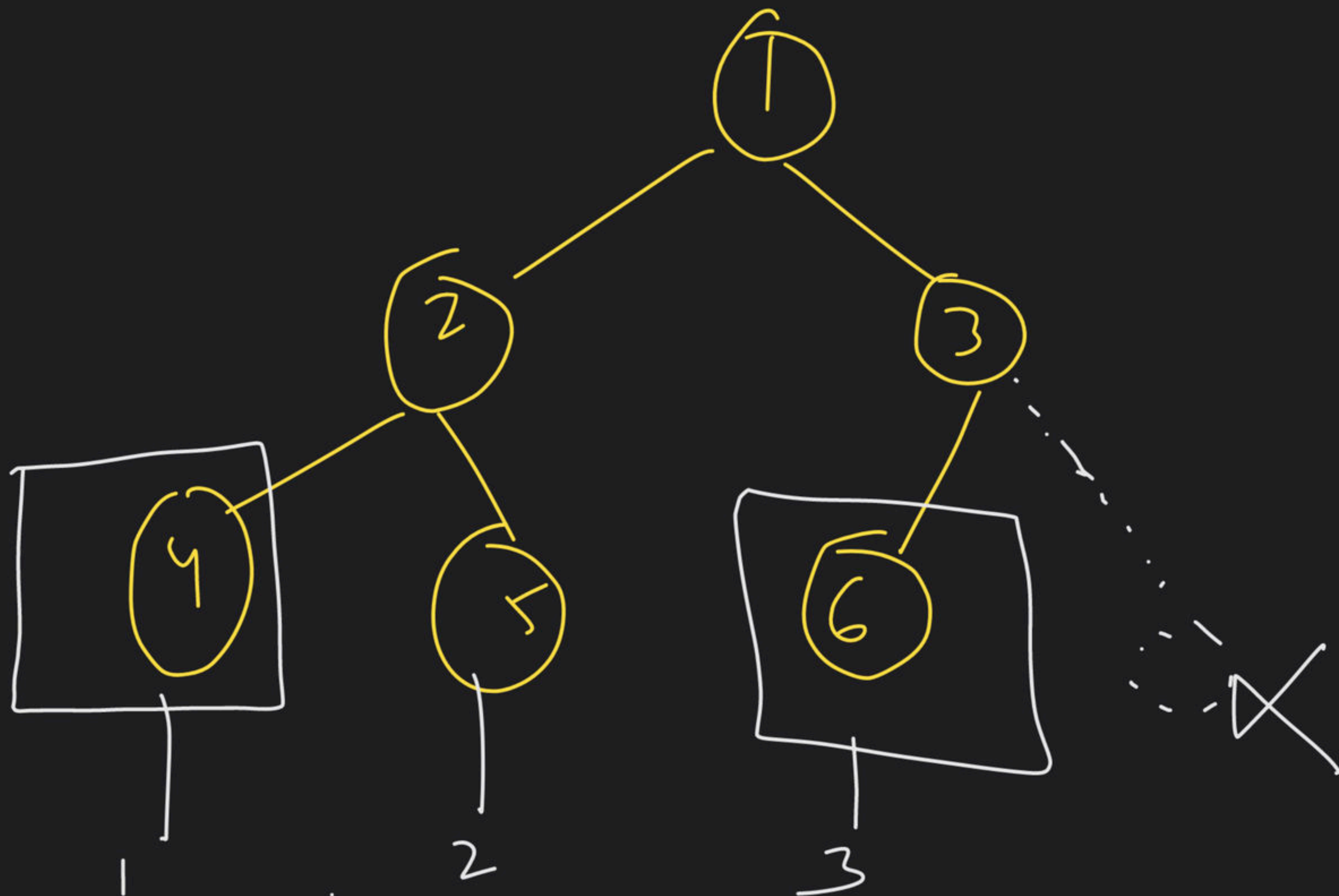
SDE → starting me → 2 → 3
scad

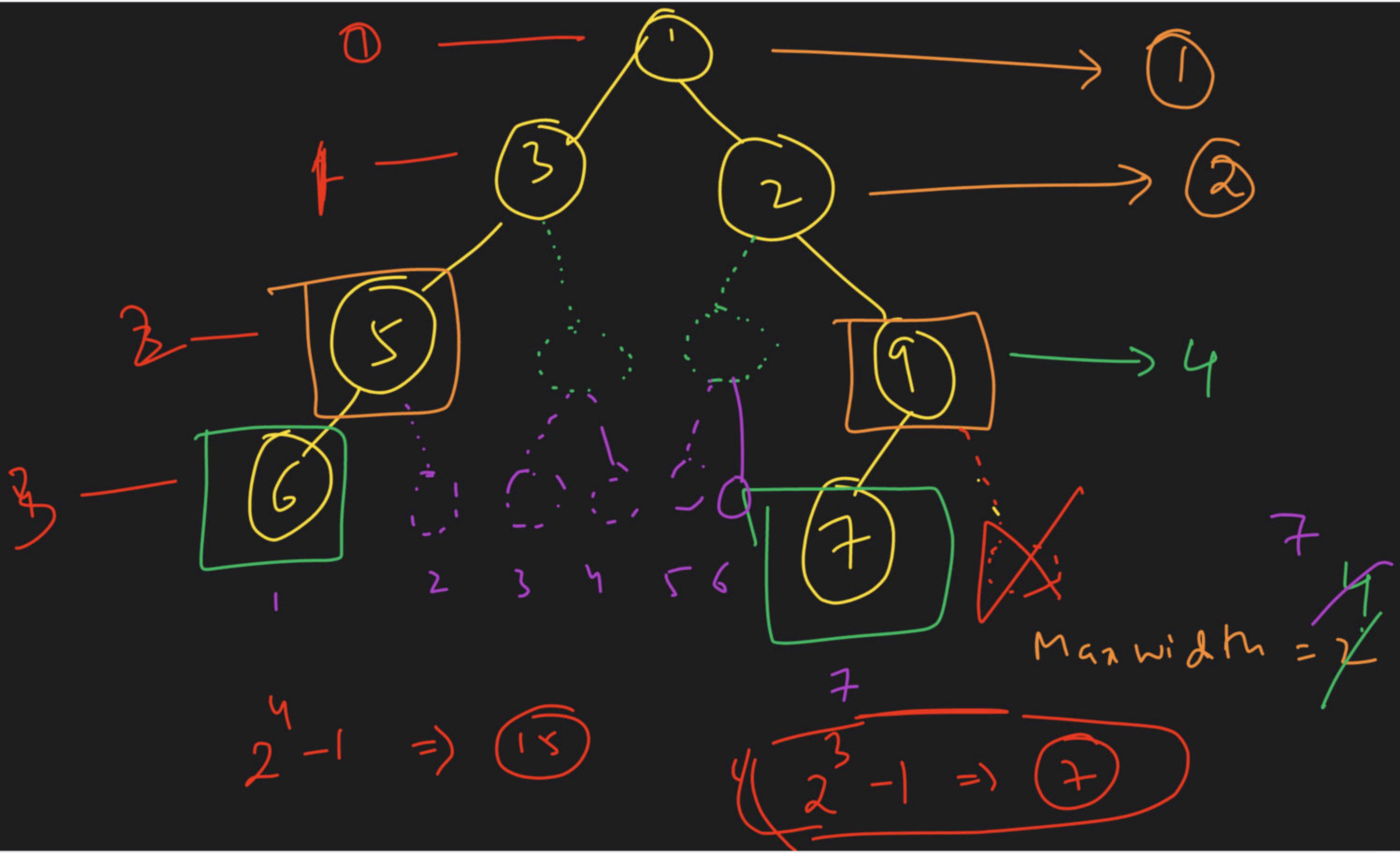
Team → switch

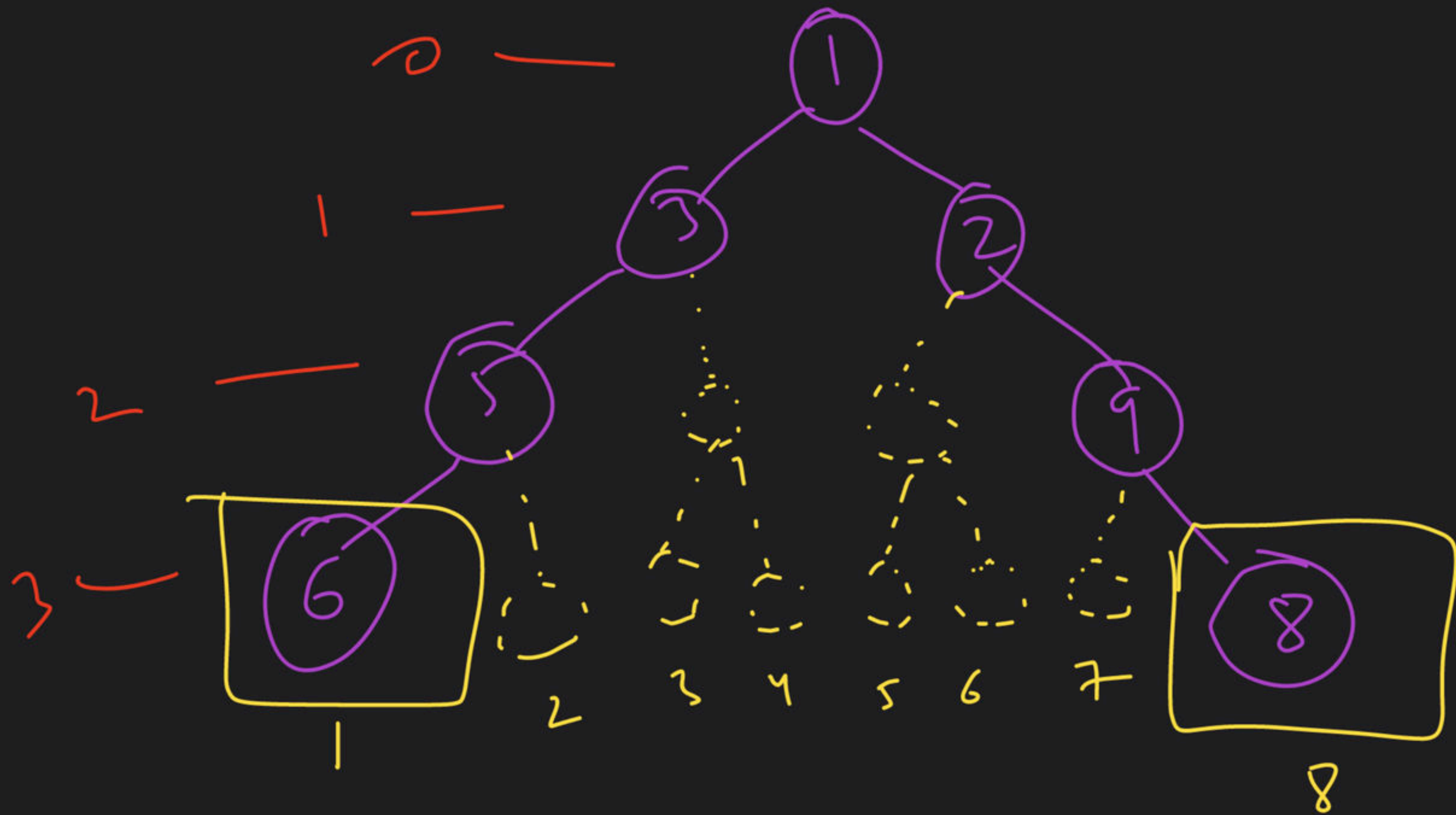
① LC 662. Max. width of BT

Think & Read → 2:50 PM



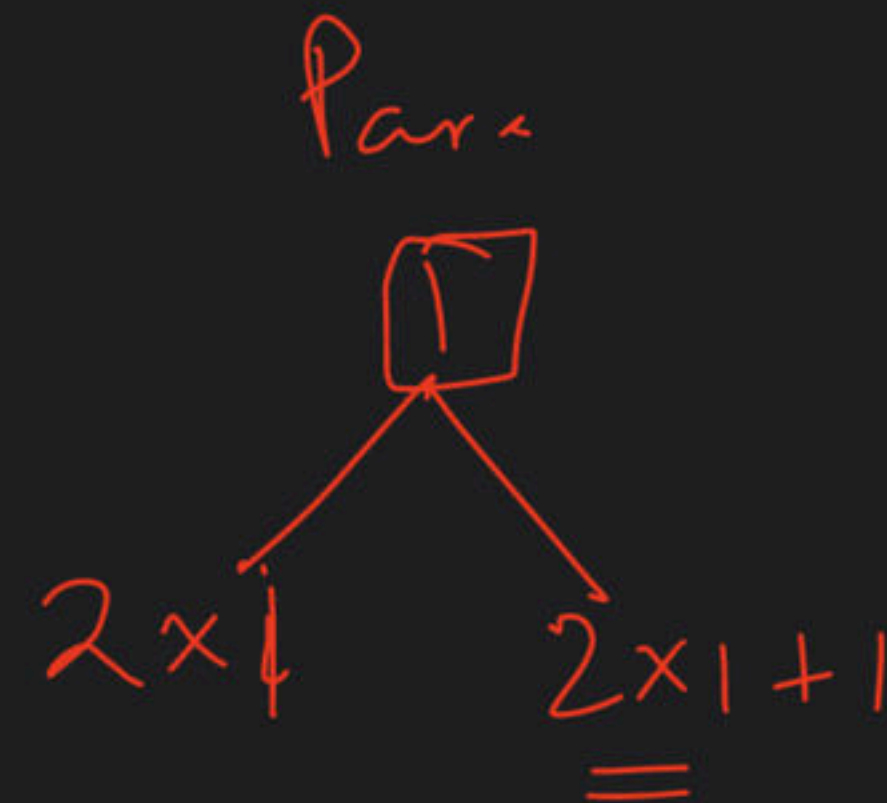
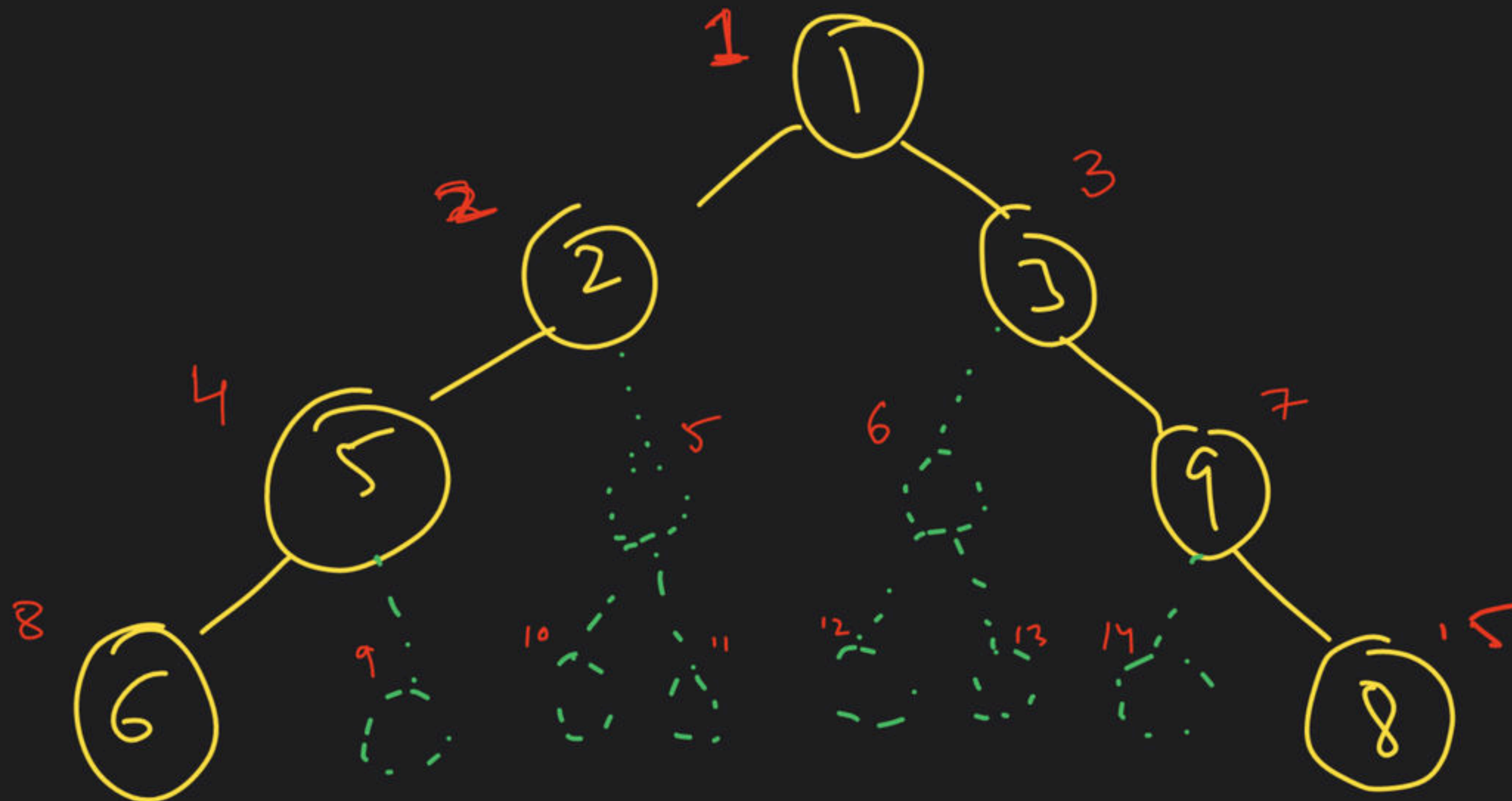






$$2^3 - 1 = 7$$

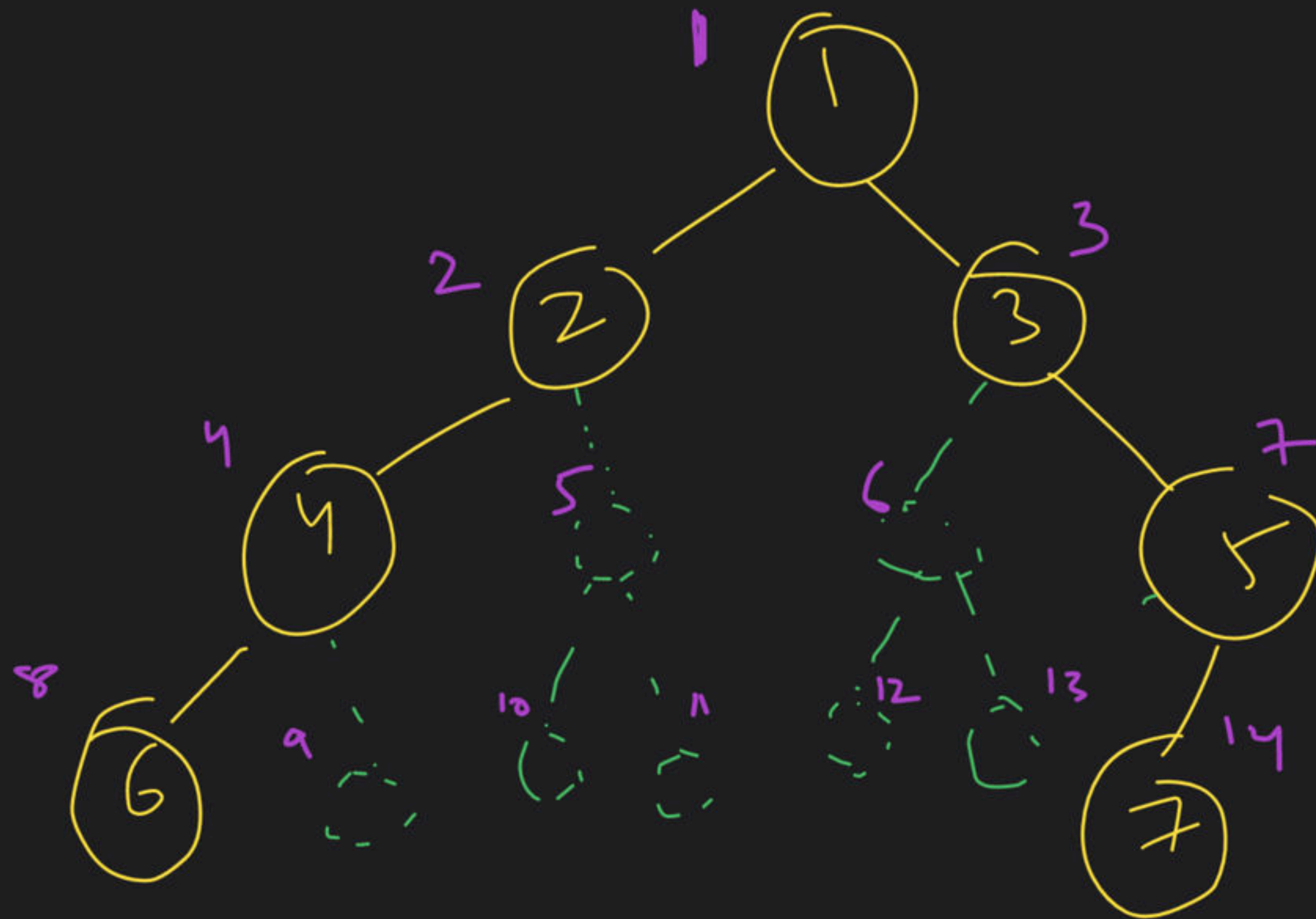




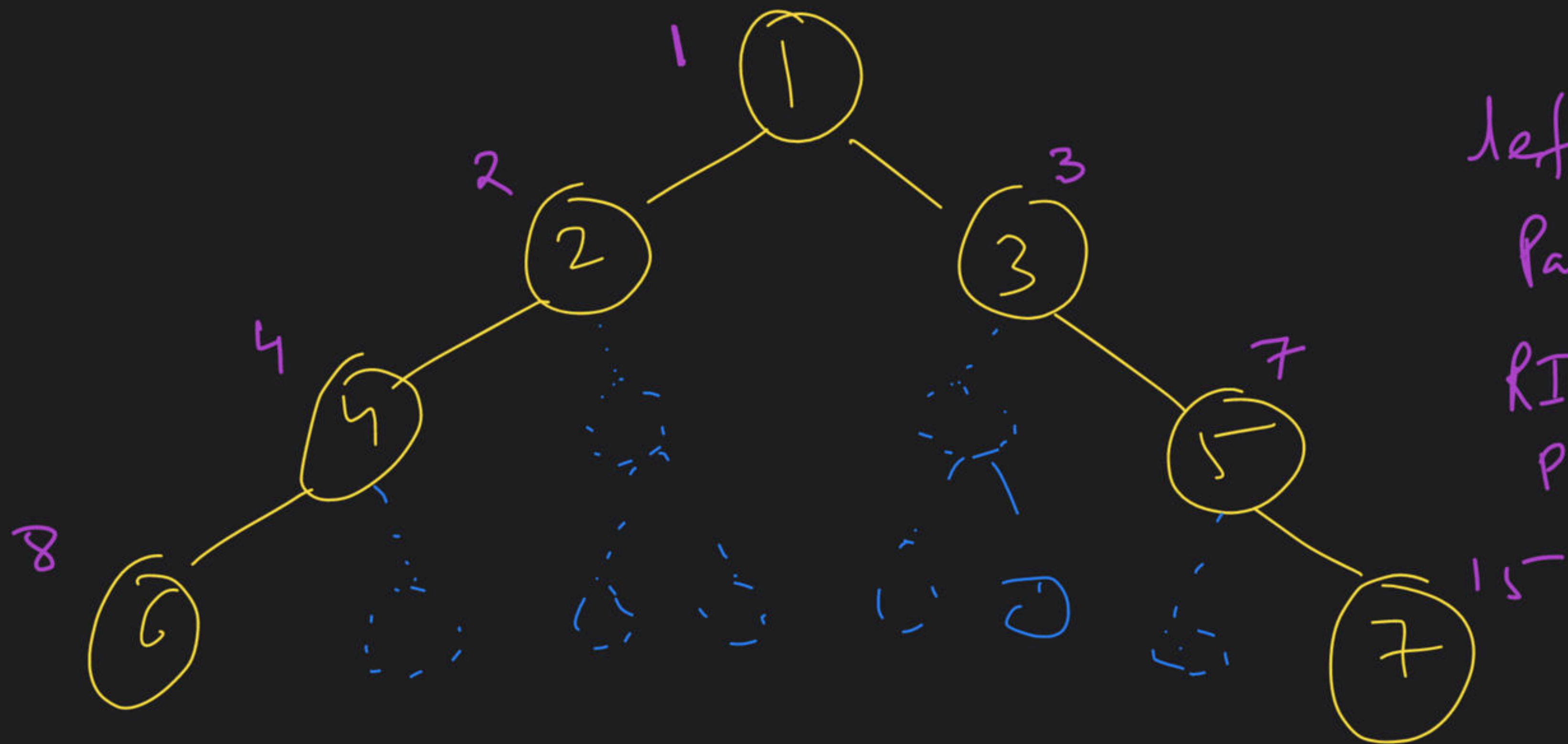
each level
width

rightmost index - leftmost index + 1

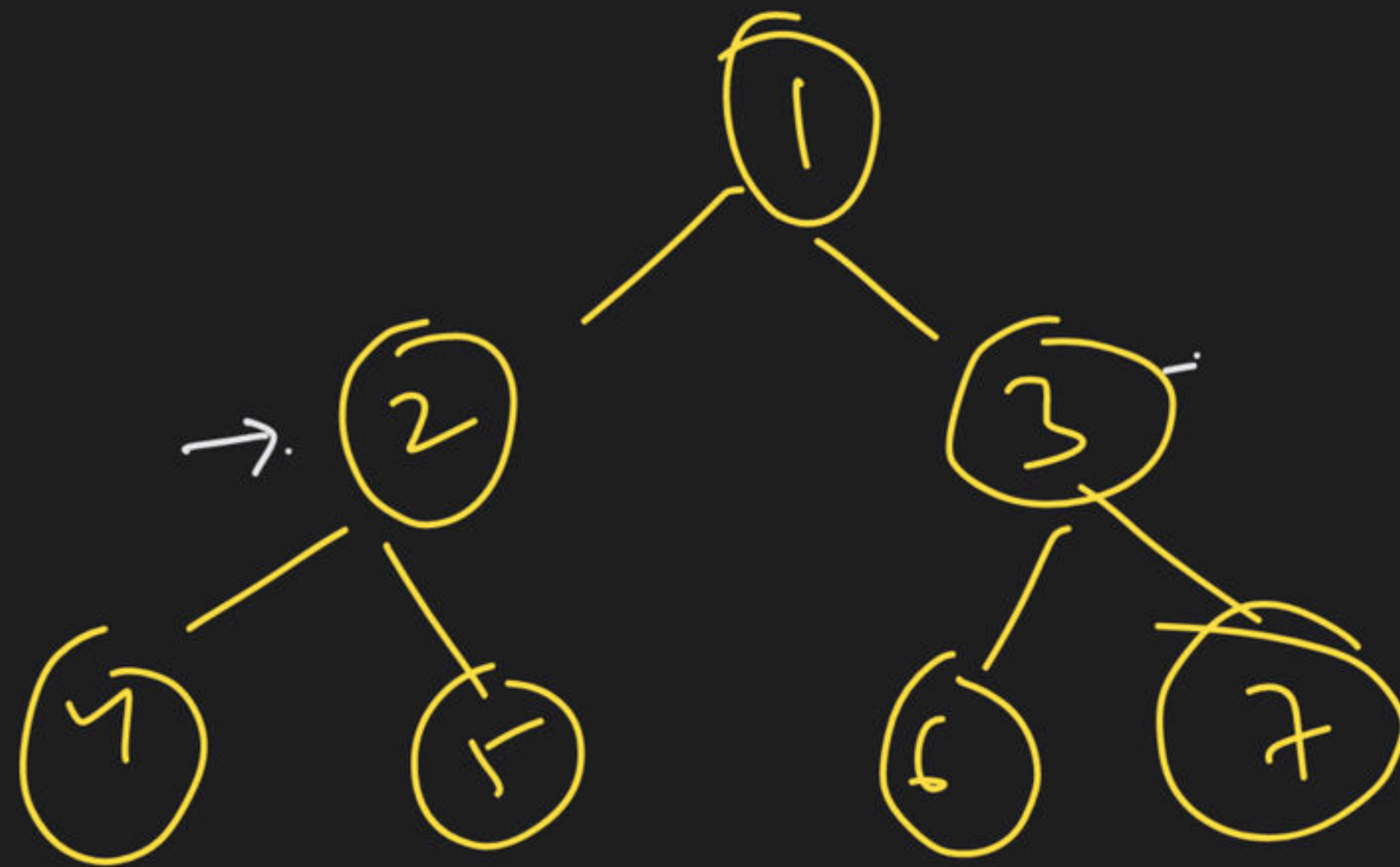
$$15 - 8 + 1 = 8$$



$$14 - 8 + 1 = 7$$



left = index
Parent index * 2
RI index
PI * 2 + 1



while !a
{
size → v → return hi
nodes

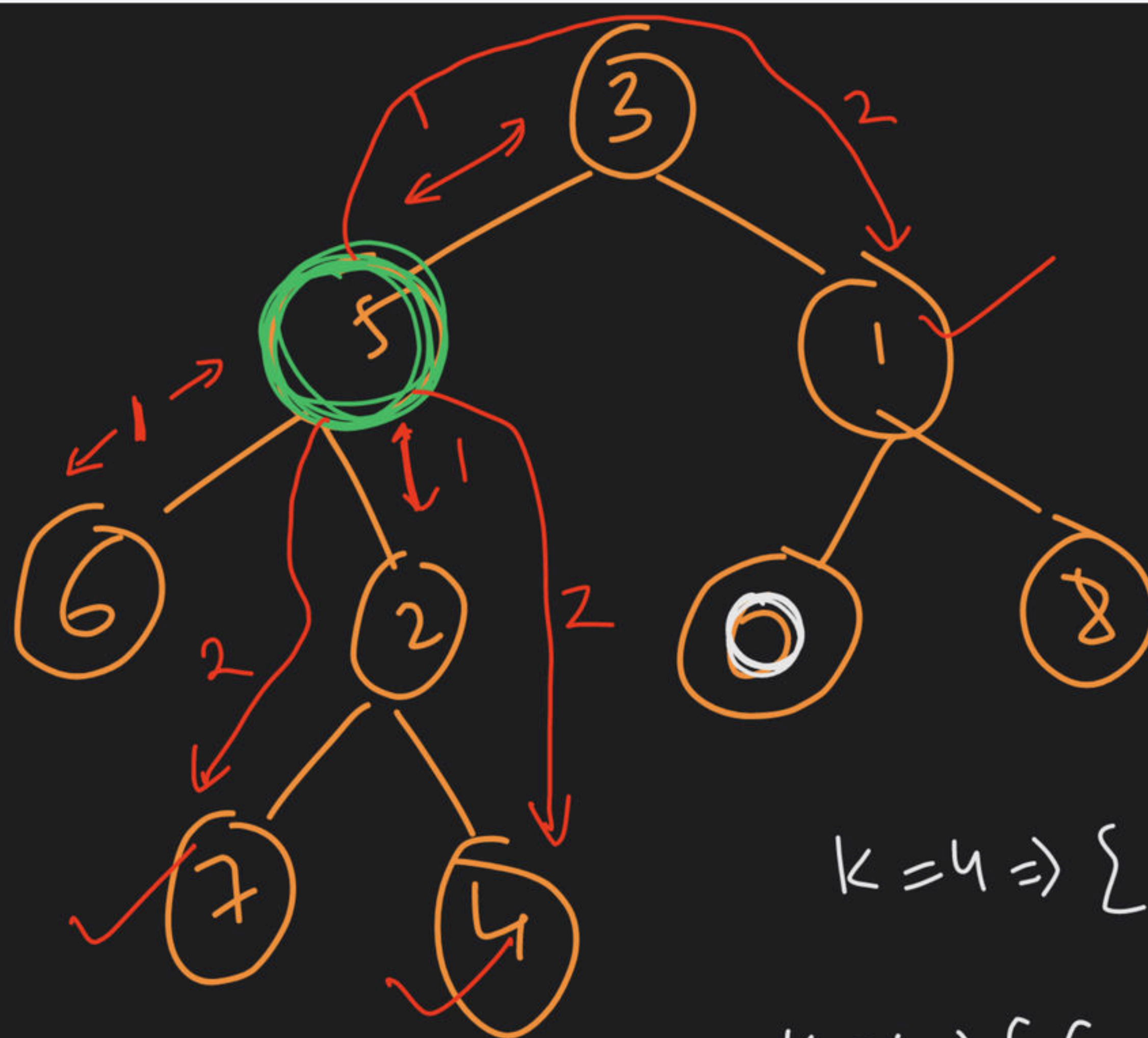
1

}

②

LC: 863, All nodes distance
k in BT

3:40 → Read 4 Think



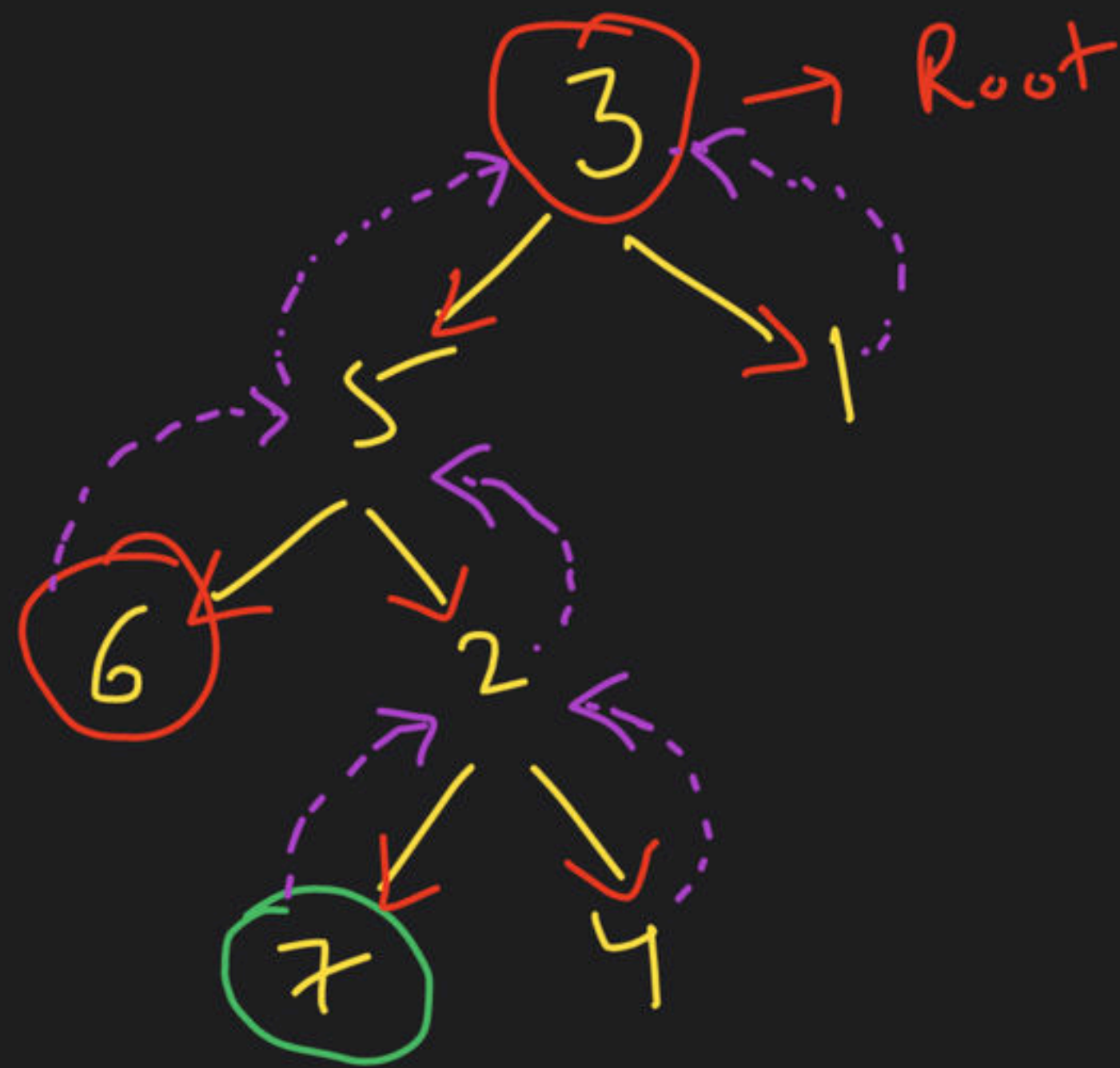
Target = 5

K = 2

$K=4 \Rightarrow \{ \}$ {7, 4, 1}

$K=1 \Rightarrow \{6, 2, 3\}$

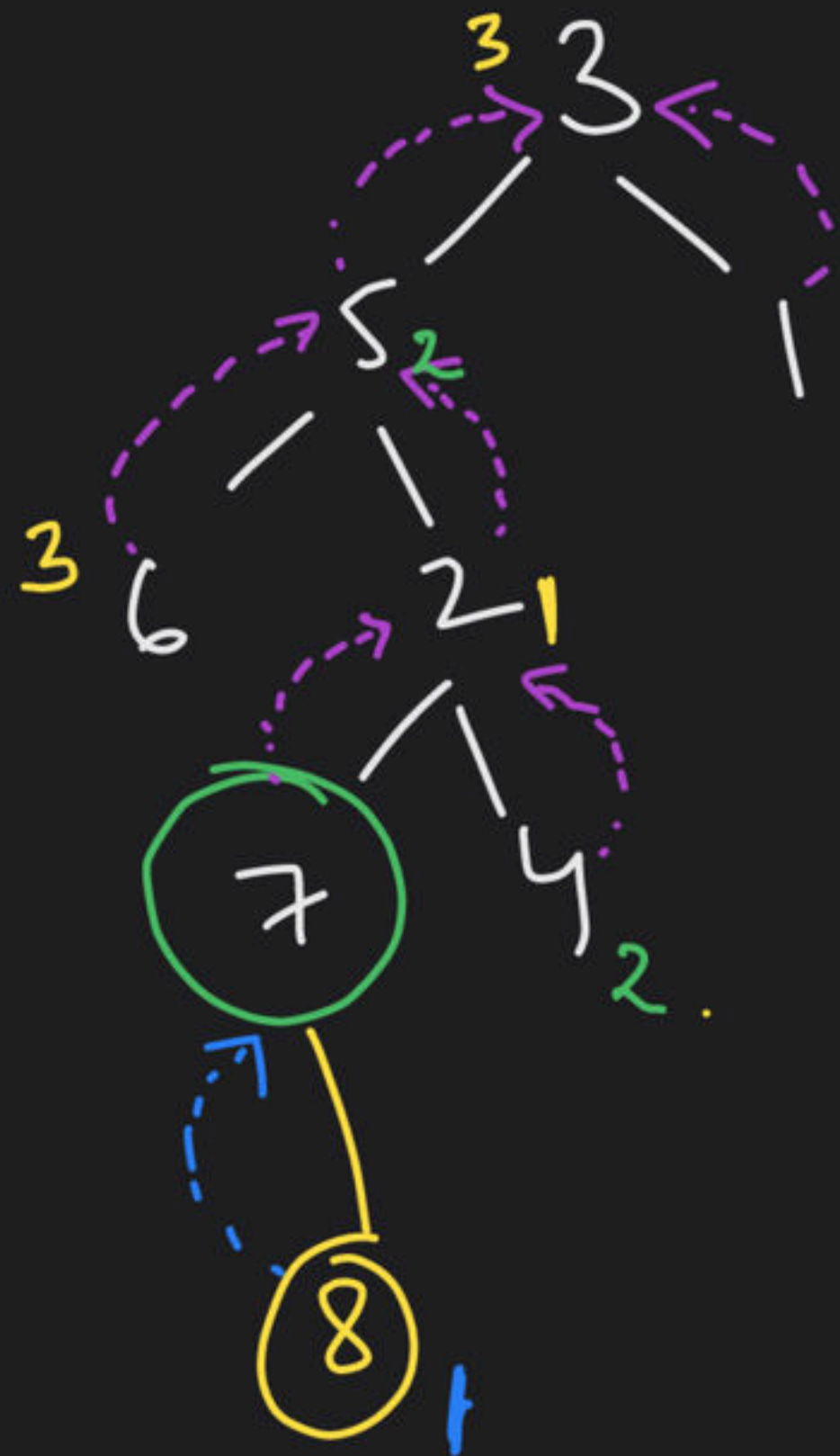
$K=3 \Rightarrow \{0, 8\}$



$k=3$
 $TN=7$

$\{6, 3\}$

Start My LOT from Target



child	parent
7	→ 2
2	→ 5
4	→ 2
6	→ 5
5	→ 3
1	→ 3
3	→ Null

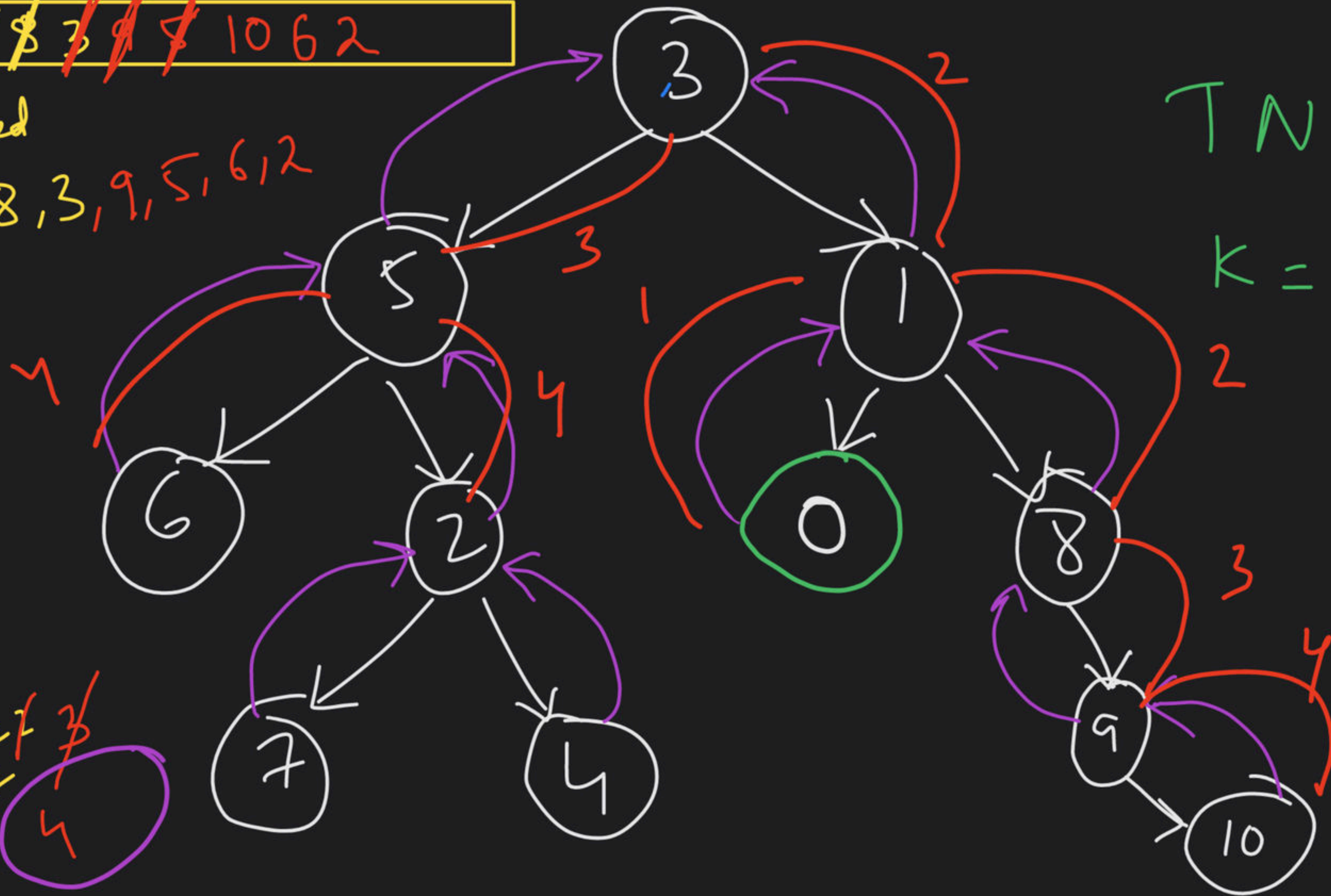
~~0~~ ~~1~~ ~~8~~ ~~3~~ ~~9~~ ~~7~~ 10 6 2

visited
1, 0, 8, 3, 9, 5, 6, 2

$TN = 0$

$K = 4$

Level 2 ~~3~~
4

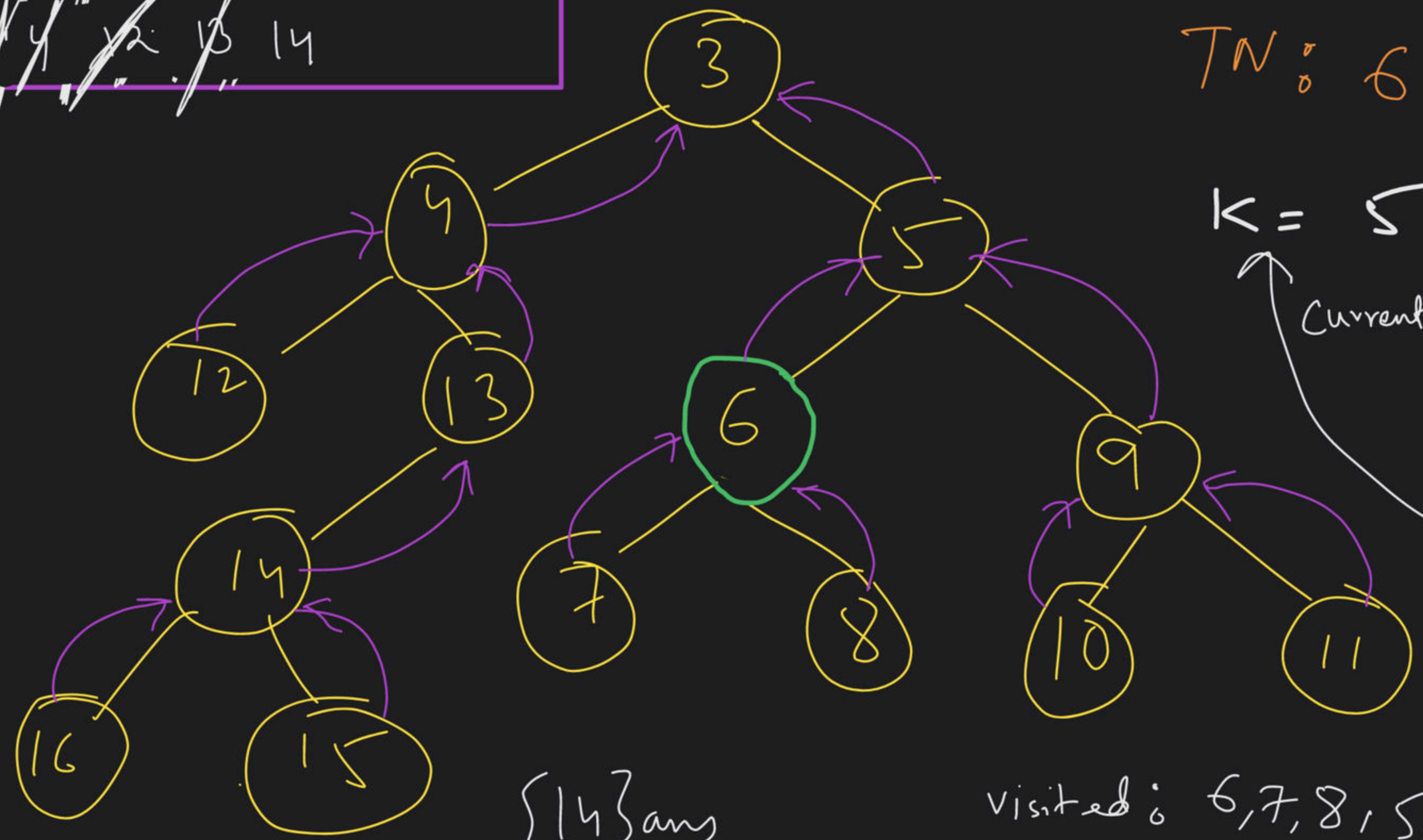


~~10~~ ~~11~~ ~~12~~ ~~13~~ ~~14~~

TN: 6

K = 5

Current level = ~~5~~
~~4~~
~~3~~
~~2~~
~~1~~



{14} ans

visited: 6, 7, 8, 5, 9, 3
10, 11, 4, 12, 13, 14

③ LL: 222, Count Complete
Tree Nodes

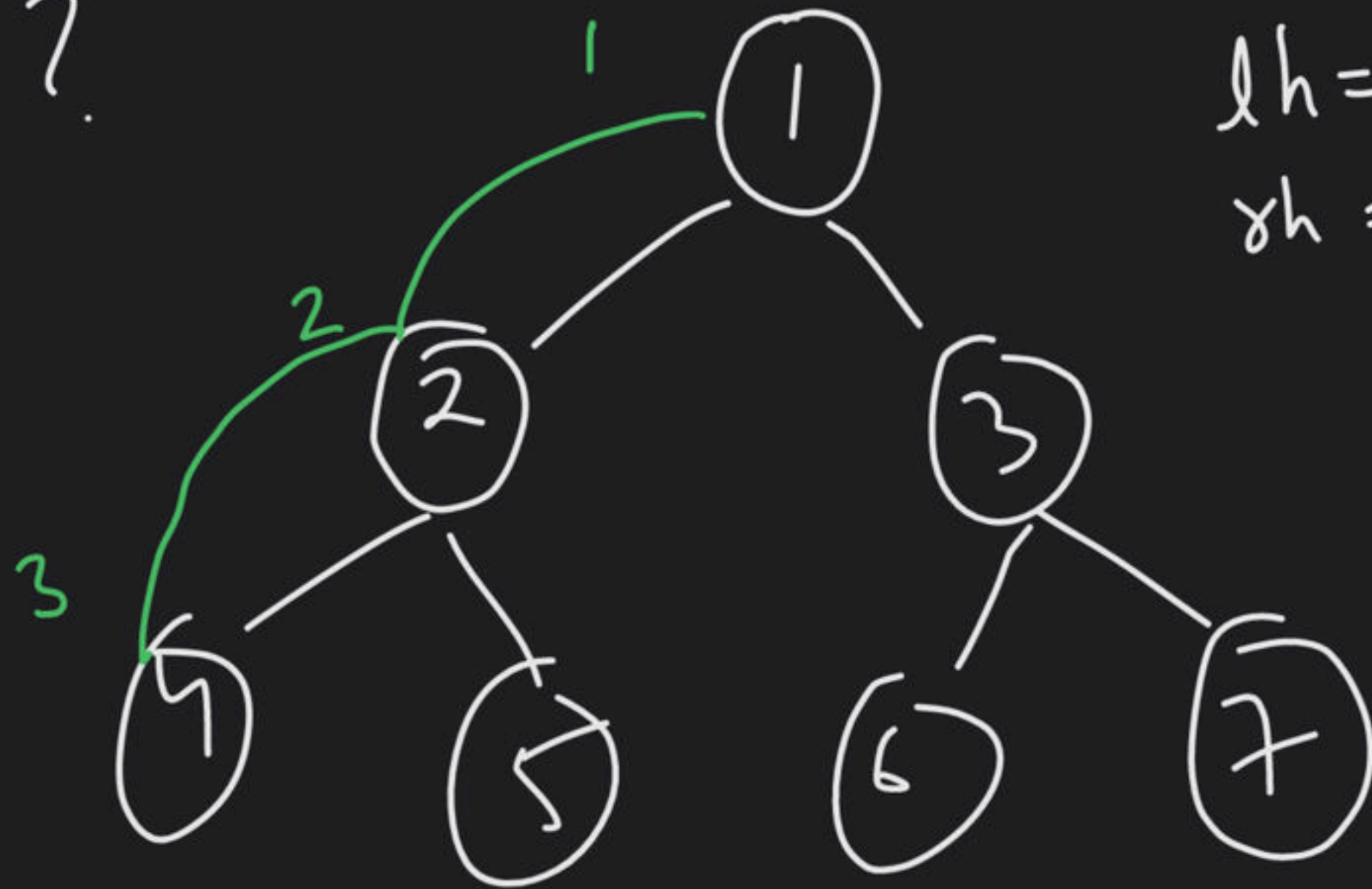
10 Break → Think & Read

Meet @ 4:50 PM ✓

① Complete BT

② $< O(n)$ count

FCBT?



lh = 3
rh = 3

N = 7

left height

$\rightarrow O(\log n)$

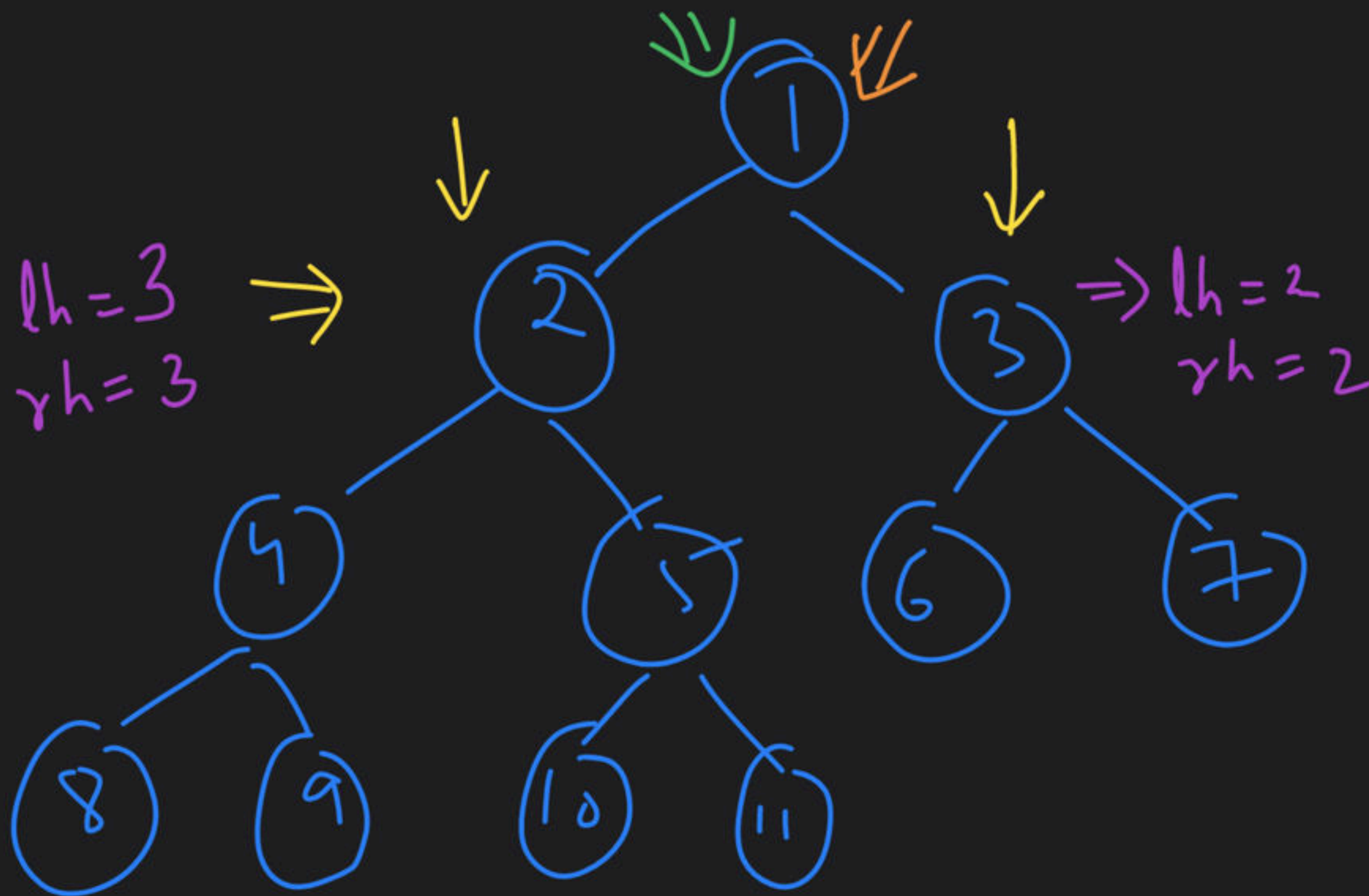
H = 3

Full CBT

$$\begin{aligned} \text{FCBT} &\Rightarrow \text{TN} \Rightarrow 2^H - 1 \\ &\Rightarrow 2^3 - 1 = 7 \end{aligned}$$

Not Full
But CBT

$lh=3$
 $rh=3$



$lh=4$
 $rh=3$ } $\log n$

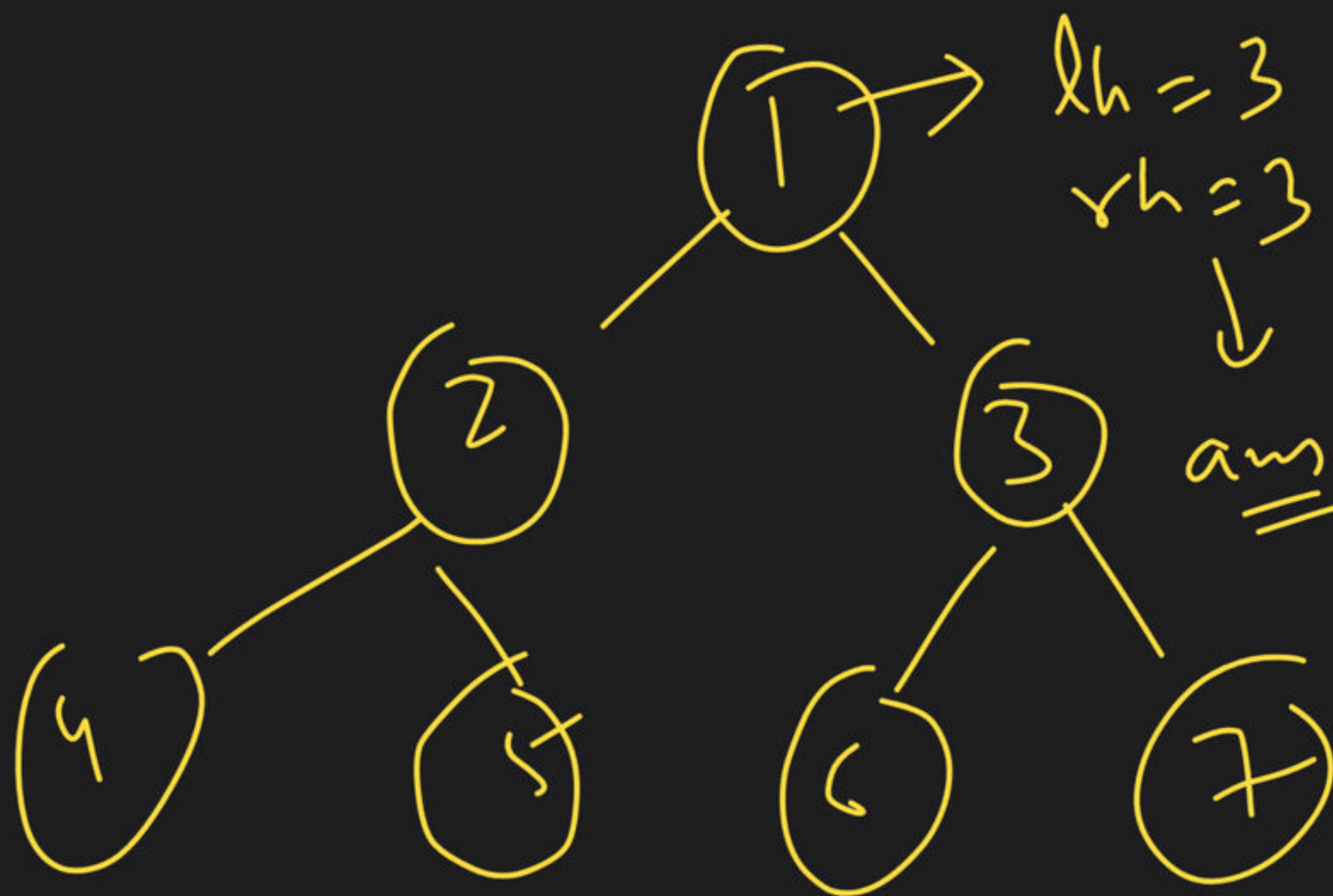
1 $\Rightarrow lh \neq rh$
Not FCBT

② $lh == rh \Rightarrow \log n$
FCBT $\Rightarrow 2^h - 1$

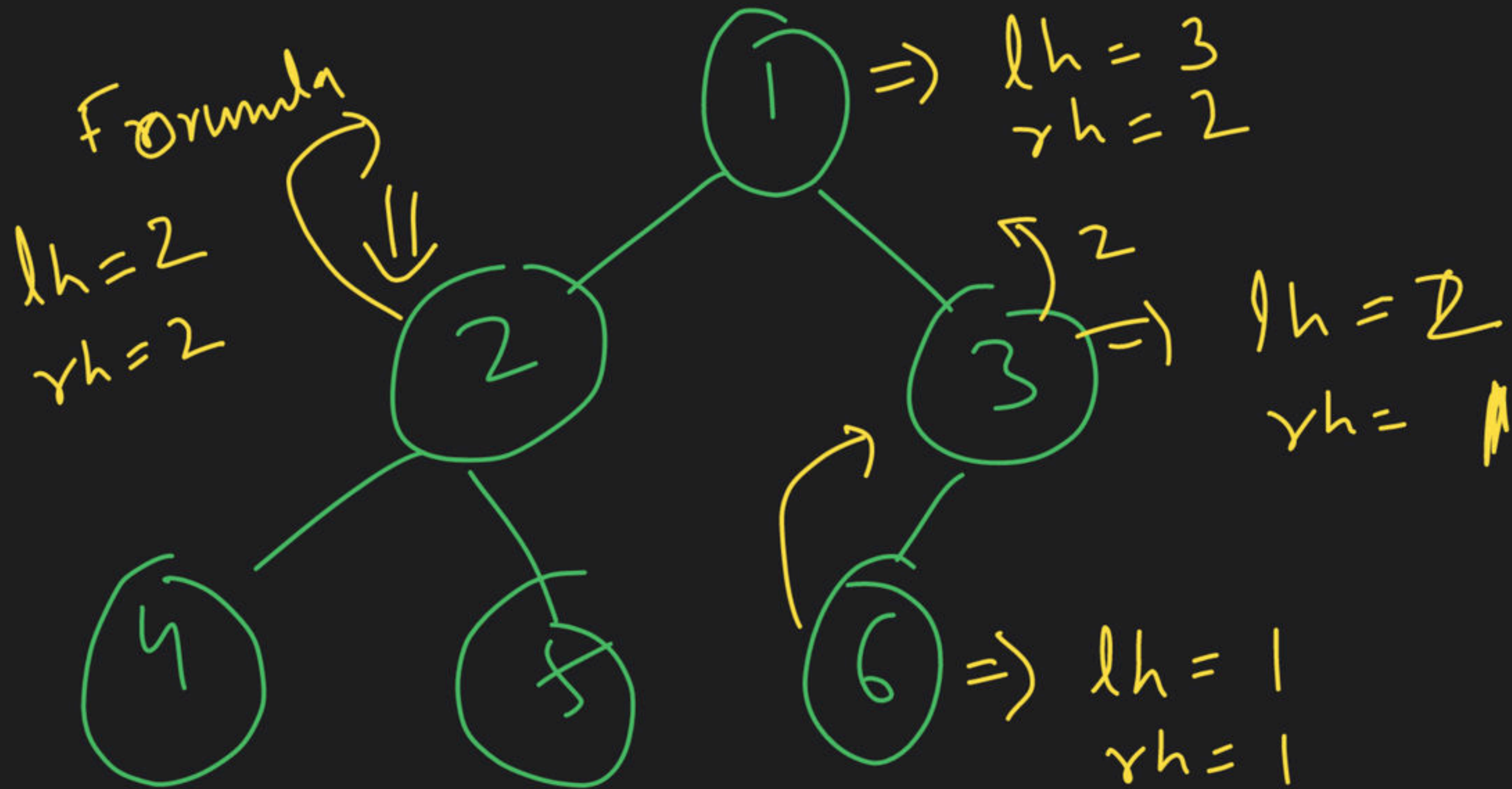
③ $lh == rh \log n$
FCBT $\Rightarrow 2^2 - 1 = 3$

$2^3 - 1 = 7$

✓✓



Best ^{log₅}
ans



$$< O(n)$$

$$lh \neq rh$$



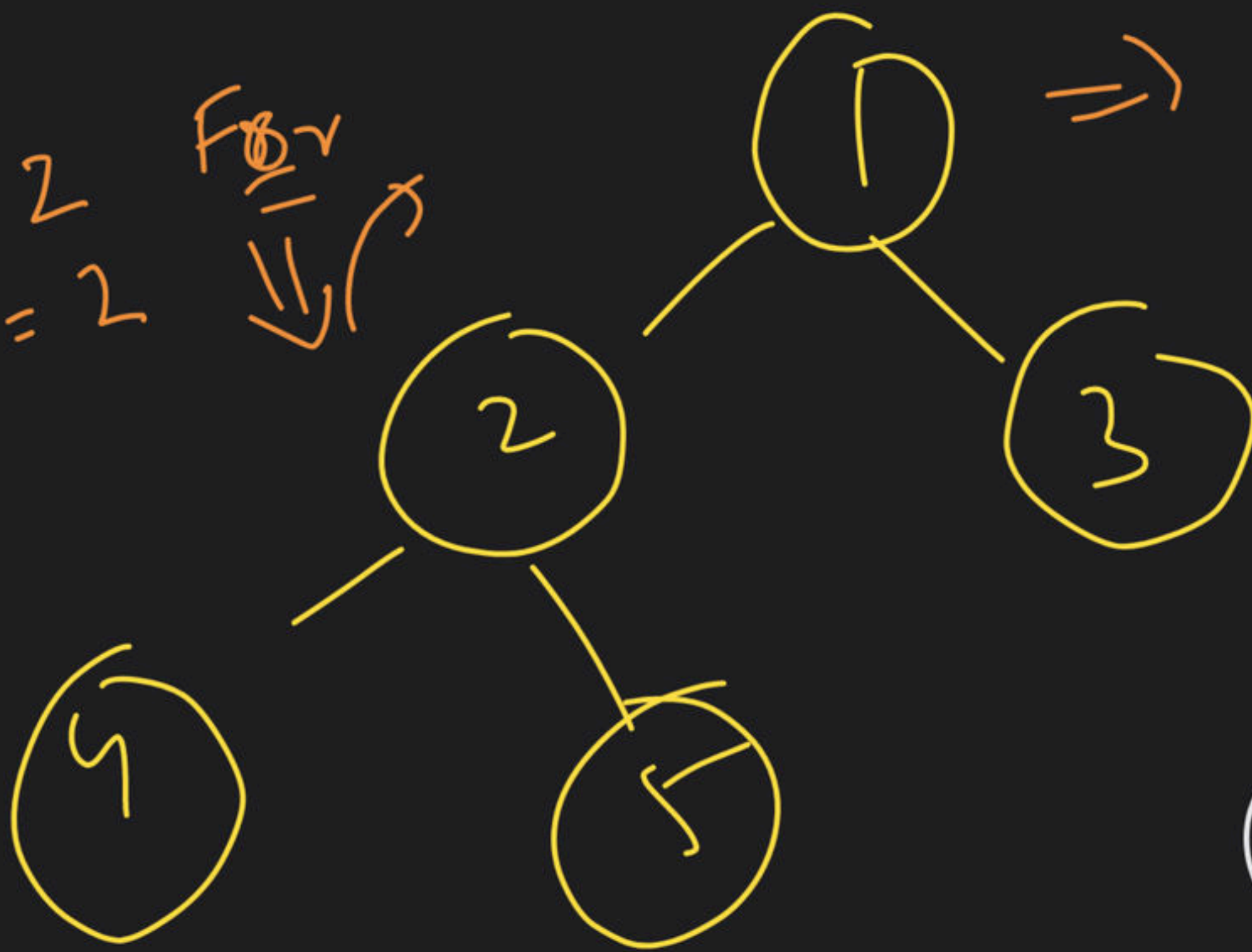
$$1 + \text{lang} + \text{ran}$$

$$1 + 1 + 0$$

$$2$$

lh = 2
rh = 2

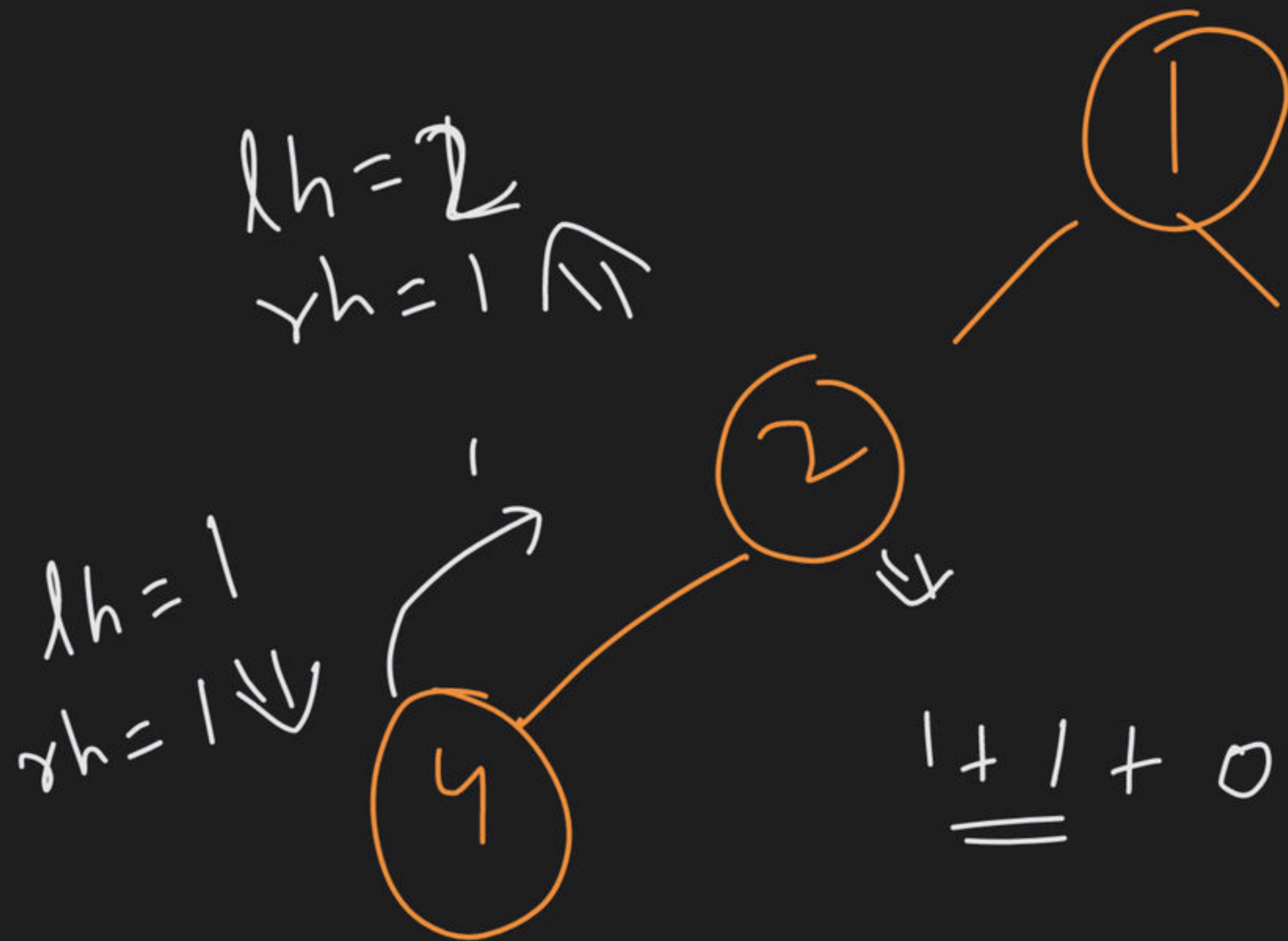
For
↓



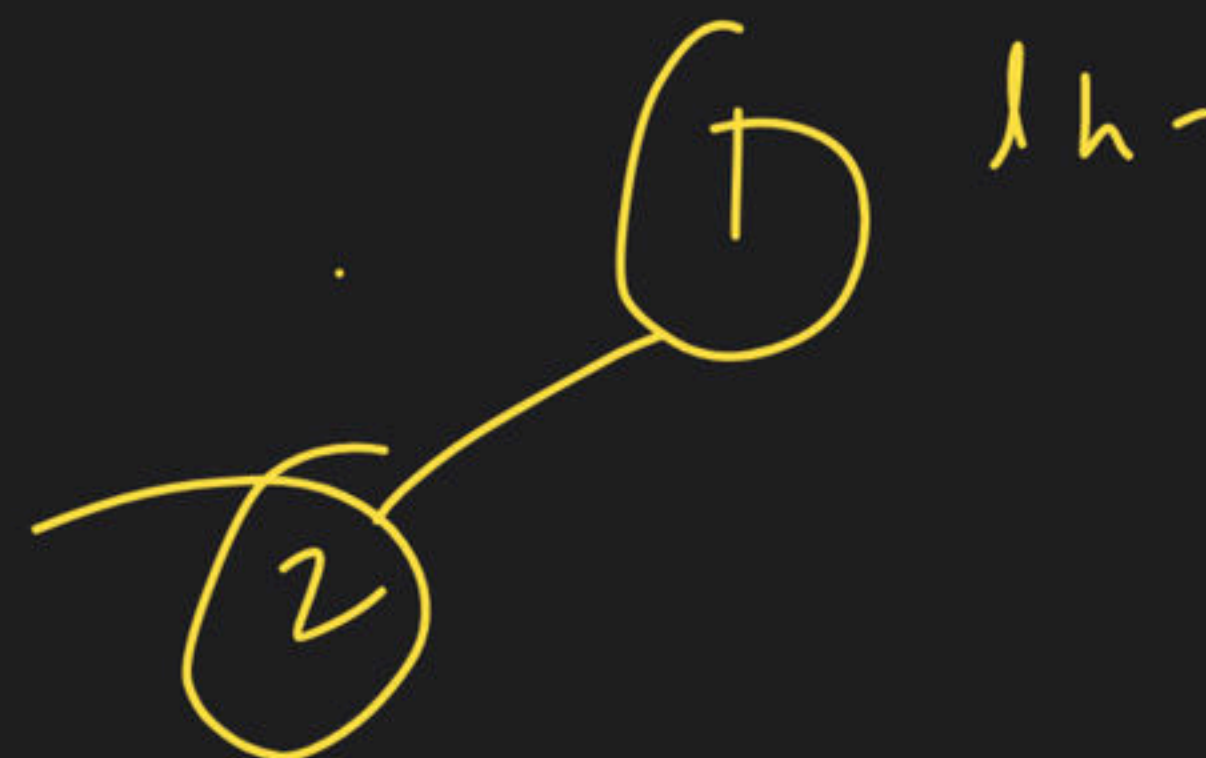
$\Rightarrow$ lh = 3
rh = 2 $\rightarrow 1 + () + ()$

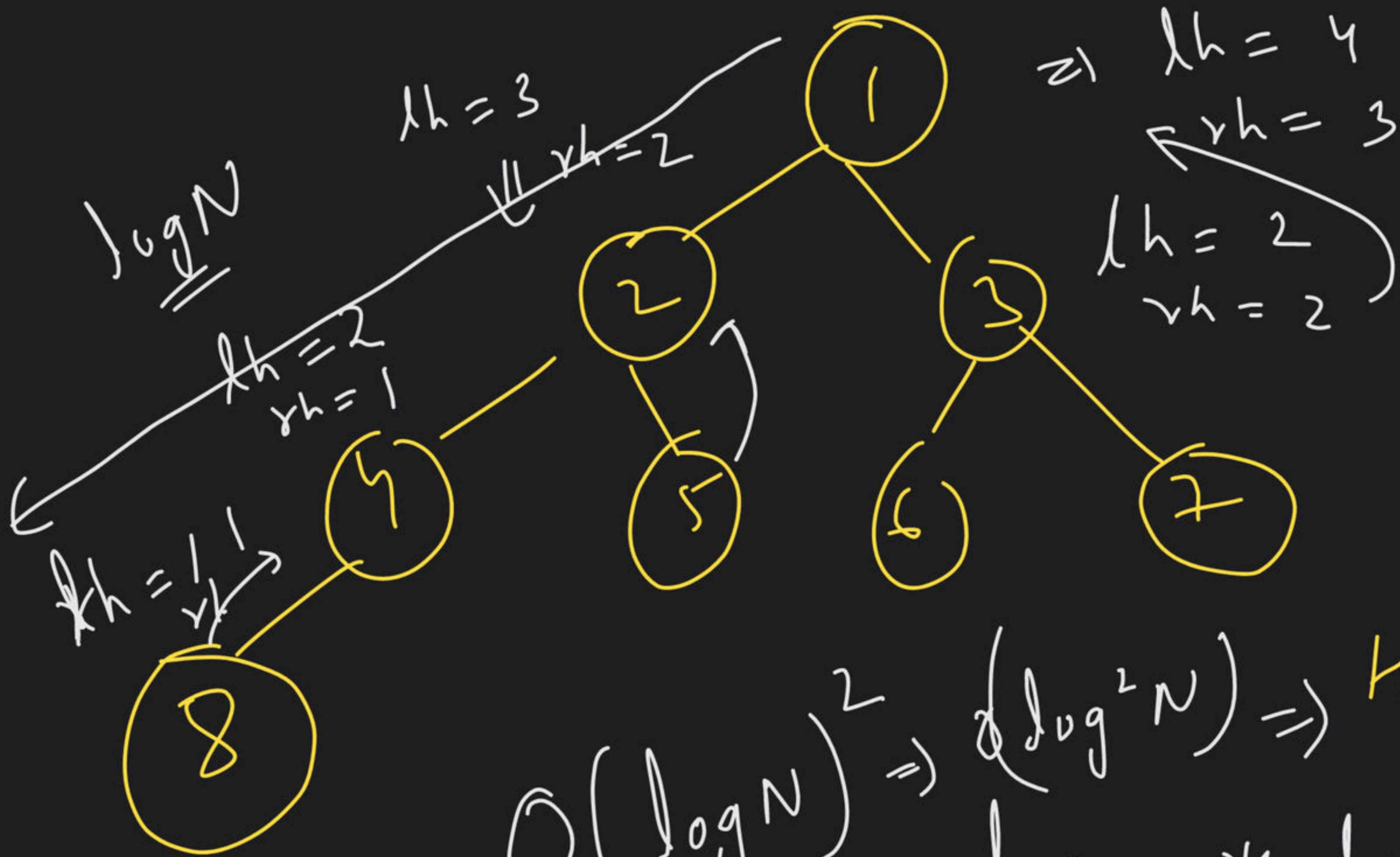
lh = 1
rh = 1

< 0/4)

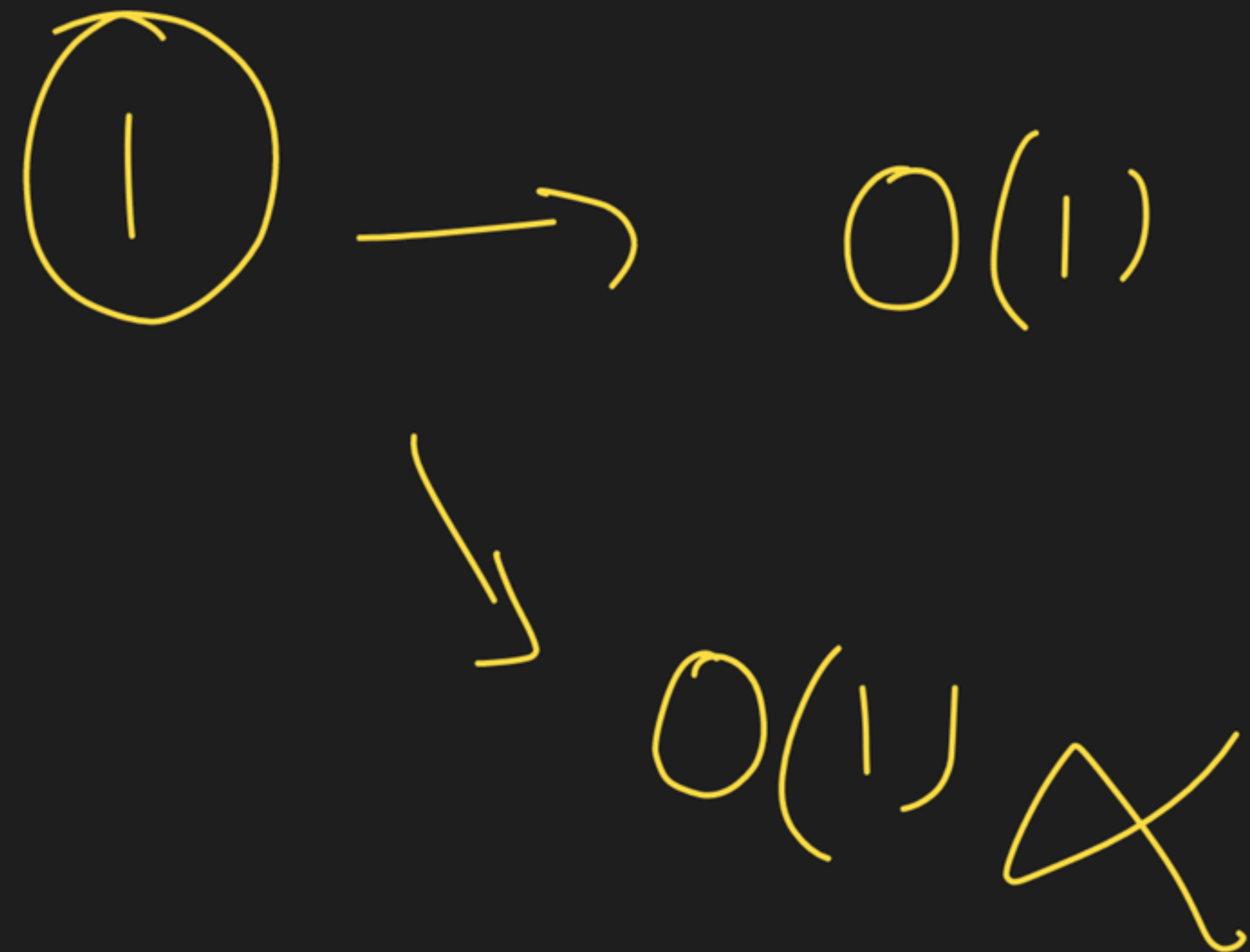


$\Rightarrow lh=3$
 $rh=2$
 $\Rightarrow$ $1 + () + ()$
 $lh=1$
 $rh=1$





$$O(\log^2 N) \Rightarrow \log n * \log n \Rightarrow H \Rightarrow \log n$$





Not a FCBT

$1 + \text{lcount} + \text{rcount}$

FCBT

FCBT

④

LC: 968 BT Cameras

⇒ → Read & Think

5:35 PM → ★

⇒ 1. W helps LRN

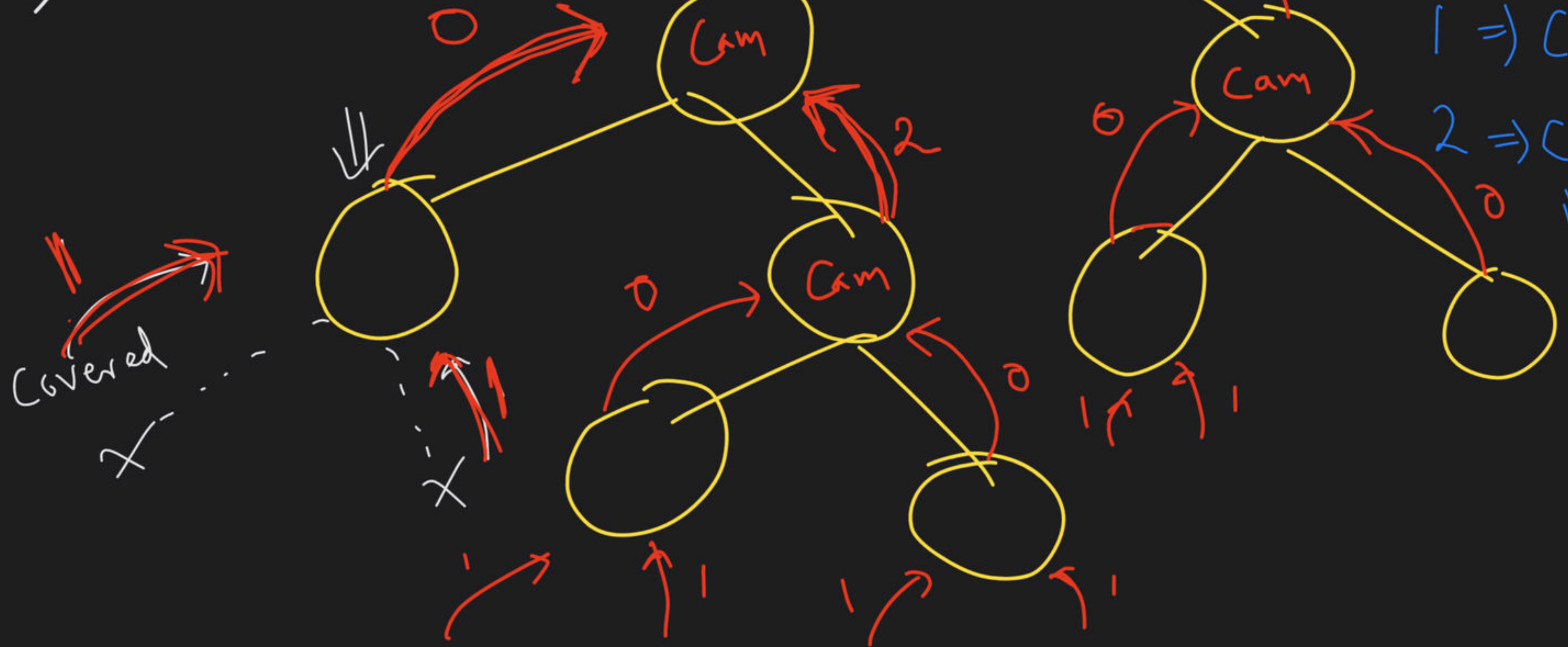
⇒

status ^{NO coverage}
0 ⇒ NO

Camera

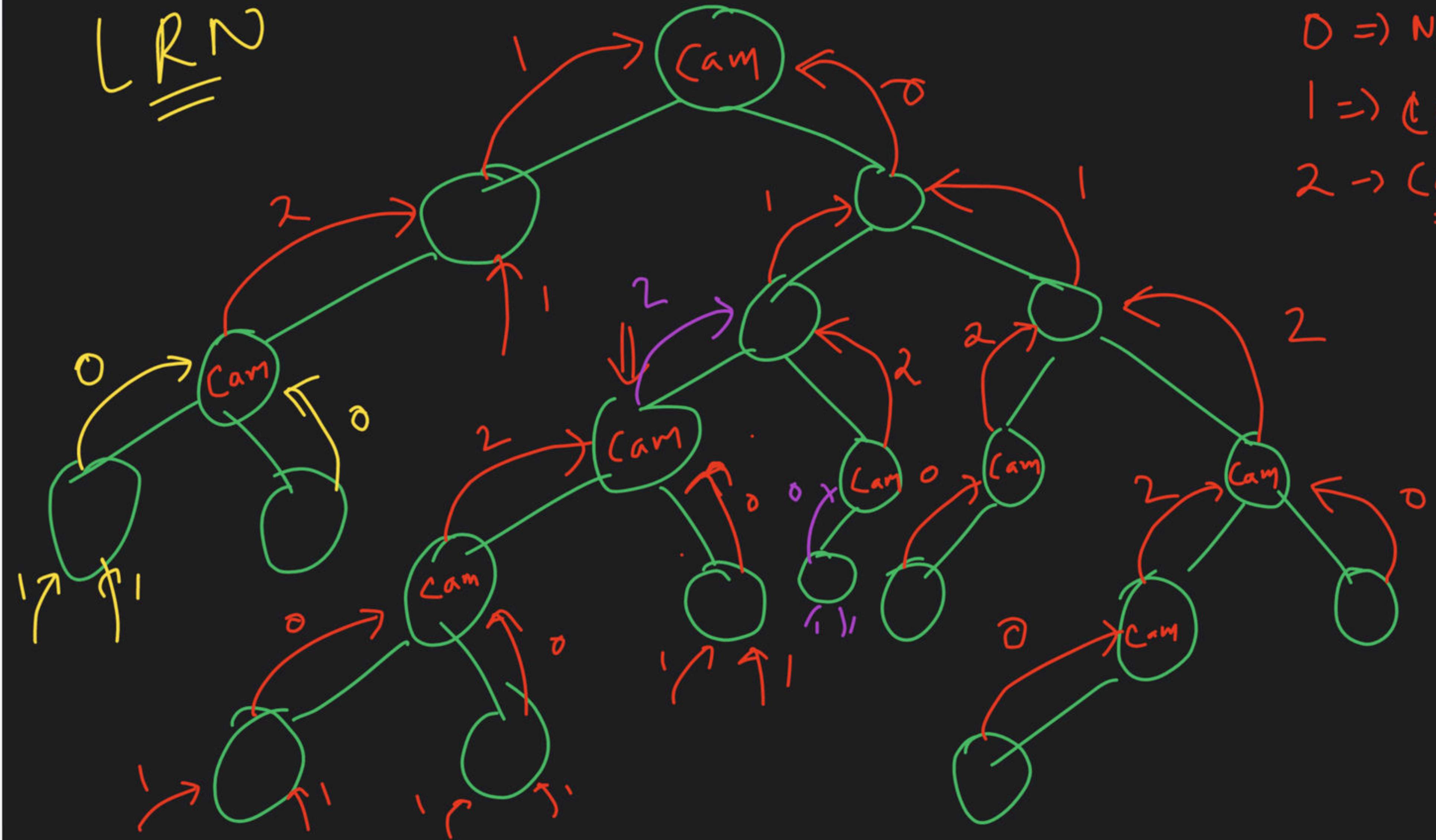
1 ⇒ Covered

2 ⇒ Camera installed =



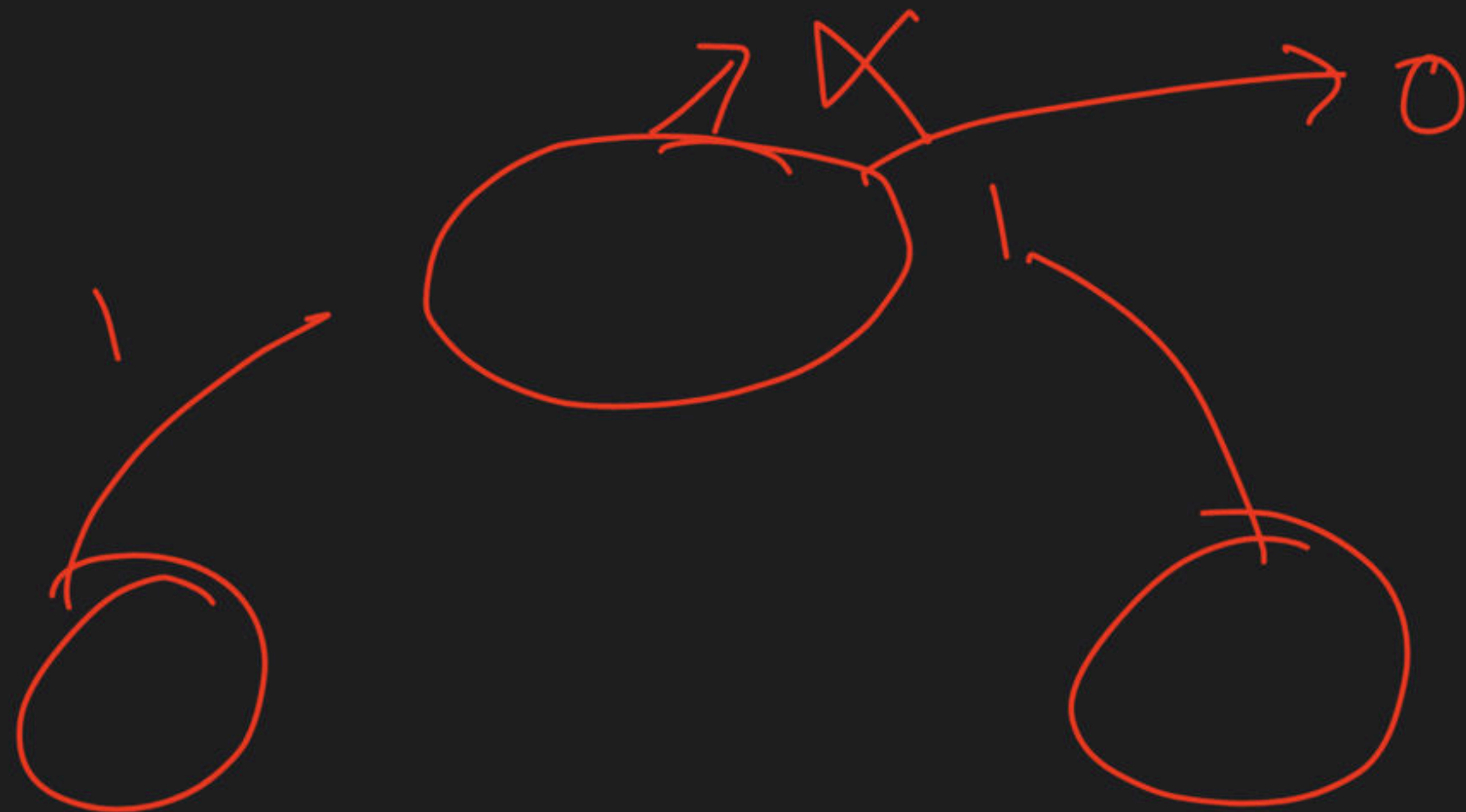
LRN

0 => Not covered
1 => Covered
2 -> Cam



```
✓✓ if (left == 0 || right == 0)
    {
        // install camera
        return 2
    }
```

```
if (left == 2 || right == 2)
    return 1;
```

LC $\Rightarrow$ 297